

Auto-Decte: Trust-Constrained Document Intelligence through Candidate–Fact Separation and Selective Prediction

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Abstract

Document intelligence in payroll-adjacent and production workflows must control not only recognition error but also how machine output becomes an authoritative record. We present Auto-Decte, a trust-constrained human-in-the-loop architecture for structured industrial forms. Its central invariant separates immutable evidence, append-only machine candidates, attributable human decisions, and versioned facts: across the exposed application paths, perception, retrieval, and language-model modules have no direct write path to the fact plane. A selective recognizer abstains when a template-distance margin is below a calibration threshold, while a review action is the only transition that can create or revise a fact. We evaluate the implementation using 4,350 deterministic synthetic image rows, disjoint calibration and evaluation seeds, seven perturbation families, five injected trust-boundary faults, and 70 automated tests. The 2,610 evaluation rows comprise 29 conditions applied to 90 seed–font–digit base units and are therefore not statistically independent. A threshold selected on 1,740 calibration rows for at most 1% empirical selective risk accepted 292 evaluation predictions (11.19% coverage) with no observed accepted error. A HOG–linear-SVM baseline achieved 99.81% accuracy at full coverage, showing that the contribution is governance of uncertain outputs rather than a new state-of-the-art recognizer. All five injected faults left facts unchanged and produced complete audit evidence. These results support the architectural mechanism while delimiting its evidence: field effectiveness still requires validation on independently transcribed production forms and human-review studies.

Keywords: document intelligence, human-in-the-loop, selective prediction,

1. Introduction

Document processing is often evaluated as a prediction problem: an image enters a model and structured values leave it. That abstraction is incomplete in payroll-adjacent and production workflows. A wrong digit may alter a wage calculation, a stale record may be mistaken for current evidence, and a plausible language-model suggestion may be granted authority that its provenance does not justify. In such settings, the consequential question is not only whether a machine output is accurate, but also whether that output can silently become an operational fact.

Modern document models substantially improve representation learning and extraction across layouts (Xu et al., 2020, 2021; Huang et al., 2022; Kim et al., 2022; Appalaraju et al., 2021; Li et al., 2023). Human-in-the-loop systems can direct difficult examples to reviewers (Budd et al., 2021; Mosqueira-Rey et al., 2023), while retrieval-augmented and generative systems can supply context or candidate values (Gao et al., 2023; Xu et al., 2024). These capabilities solve complementary inference and interaction problems. They do not, by themselves, constrain the authority of a machine output in the application data model. A high-scoring prediction and a retrieved historical value remain unsafe if either has an unreviewed write path to an authoritative record.

We study the following question: *How can machine perception and optional generative assistance support structured-form digitization while making human authorization an explicit, auditable prerequisite for fact creation?* Auto-Decte addresses this question through two coupled mechanisms. First, *candidate-fact separation* places perception results, retrieved references, and language-model suggestions outside the authoritative fact plane. Second, *selective prediction* rejects low-margin recognition results and exposes abstention as an ordinary workflow outcome rather than an exception (Chow, 1970; Geifman and El-Yaniv, 2017, 2019).

The resulting claim is deliberately narrower than “safe AI.” Across the application paths exercised in this work, machine modules cannot directly create or revise facts. This architectural property does not guarantee that a human reviewer will detect every error, nor does it establish safety under arbitrary database compromise or malicious operating-system access. It makes

each transition into the fact plane explicit, attributable, and traceable to evidence.

This paper makes four contributions:

1. We formalize a four-plane trust model—evidence, candidates, human decisions, and versioned facts—with a machine-to-fact noninterference invariant scoped to exposed application services.
2. We instantiate the model in a runnable Python 3.11 and Streamlit system for fixed industrial forms, combining QR template identity, ArUco localization, selective cell recognition, optional AI review, local retrieval, and auditable export.
3. We provide a deterministic evaluation package covering 4,350 synthetic image rows, calibration/evaluation seed separation, seven perturbation families, five fault injections, resilience checks, and 70 automated tests.
4. We report both favorable and unfavorable evidence. Risk-targeted abstention produced no observed accepted error at low coverage, while a HOG–linear-SVM baseline substantially outperformed the built-in template recognizer at full coverage. The architecture, not recognition novelty, is therefore the main contribution.

The remainder of the paper distinguishes related inference and governance work (Section 2), defines the trust invariant (Section 3), describes the implementation and selective rule (Section 4), and presents a pre-registered-by-code synthetic protocol and results (Sections 5–6). Sections 7–9 delimit industrial relevance and remaining validation needs.

2. Related work

2.1. Document understanding and structured forms

Document layout analysis has progressed from engineered segmentation pipelines to multimodal representation learning (Binmakhashen and Mahmoud, 2019). LayoutLM and its successors jointly model textual, visual, and spatial signals (Xu et al., 2020, 2021; Huang et al., 2022); DocFormer fuses text, vision, and spatial features (Appalaraju et al., 2021); Donut removes a separate OCR stage (Kim et al., 2022); and TrOCR casts text recognition as sequence modeling (Li et al., 2023). PP-OCrv3 represents a highly optimized lightweight OCR pipeline (Li et al., 2022). Recent surveys show that form understanding spans key–value extraction, table structure, handwriting, and heterogeneous layouts (Abdallah et al., 2024). Auto-Decte occupies a

narrower regime: fixed, versioned forms whose geometry is known in advance. This restriction permits deterministic localization and makes governance, abstention, and traceability the research focus.

Fiducial markers provide a reproducible geometric reference for this regime. ArUco offers efficient square-marker detection (Romero-Ramirez et al., 2018), and optimized dictionaries increase separation among marker codes (Garrido-Jurado et al., 2016). We use QR identity and four ArUco corners to select a template and normalize perspective; we do not claim a new marker detector.

2.2. Human oversight and selective prediction

Human-in-the-loop machine learning commonly allocates labeling or correction effort where it is most informative or where models are uncertain (Budd et al., 2021; Mosqueira-Rey et al., 2023). Human–AI interaction guidance additionally stresses communicating uncertainty, supporting correction, and preserving user control (Amershi et al., 2019). Auto-Decte adopts these interaction principles but gives review a stronger data-model role: an attributable review event is the only application transition that creates a new fact version.

Reject-option classification dates to the formal trade-off between recognition error and rejection (Chow, 1970). Contemporary selective classifiers jointly optimize prediction and coverage (Geifman and El-Yaniv, 2017, 2019); calibration and conformal methods provide related tools for interpreting or controlling uncertainty (Guo et al., 2017; Angelopoulos and Bates, 2023). We use a simple margin score because it is transparent and reproducible for fixed digit cells. The novelty claim is not the score itself, but its integration with a fact-authority boundary: both accepted and abstained recognition outcomes remain candidates until reviewed.

2.3. Provenance, generative assistance, and authority

Data provenance explains why and where a result arose (Buneman et al., 2001; Herschel et al., 2017); the W3C PROV data model provides interoperable concepts for entities, activities, and agents (World Wide Web Consortium, 2013). Auto-Decte applies the same concern operationally by linking evidence hashes, recognition attempts, review events, record versions, and export batches.

Large language models can generate structured extraction candidates (Xu et al., 2024), and retrieval-augmented generation can ground responses in

external material (Gao et al., 2023). Yet grounding does not confer authority: a relevant record may be stale, and a plausible suggestion may still be wrong. We therefore treat retrieval as read-only context and AI output as an append-only suggestion requiring human confirmation. This follows human-centered principles of reliability and control (Shneiderman, 2020) and addresses one point in the broader lifecycle of machine-learning harm (Suresh and Guttag, 2021). The NIST AI Risk Management Framework likewise emphasizes governance and traceability rather than accuracy alone (Tabassi, 2023). Auto-Decte contributes a concrete, testable application invariant for that governance layer.

2.4. Positioning of the contribution

Prior work generally optimizes one of three layers: document inference, uncertainty-aware routing, or human interaction. Auto-Decte connects them through an authority model. Machine output is useful but non-authoritative; review is not merely a correction interface but a state transition; provenance is not merely metadata but the basis of reverse trace. This positioning also explains why a stronger classifier does not invalidate the contribution: any recognizer can replace the current baseline without changing the candidate-fact contract.

3. Trust model and problem formulation

Let a captured form image be evidence $e \in E$, and let M denote the set of machine modules exposed through the application: perception, optional language-model review, and retrieval. A module may derive a candidate $c \in C$ from evidence or existing non-authoritative context. A human decision $h \in H$ records an actor, an action, an expected record version, a reason, and referenced evidence. Authoritative state is represented as a sequence of fact versions $F_0, F_1, \dots$

The allowed transitions are

$$E \xrightarrow{m} C, \quad C \xrightarrow{\text{review}} H, \quad F_{t+1} = g(F_t, H_t), \quad (1)$$

where g validates the expected version and appends, rather than overwrites, the resulting record. Retrieval can add a reference to C but cannot supply H . The central invariant is

$$\forall m \in M, \quad m \not\rightarrow F. \quad (2)$$

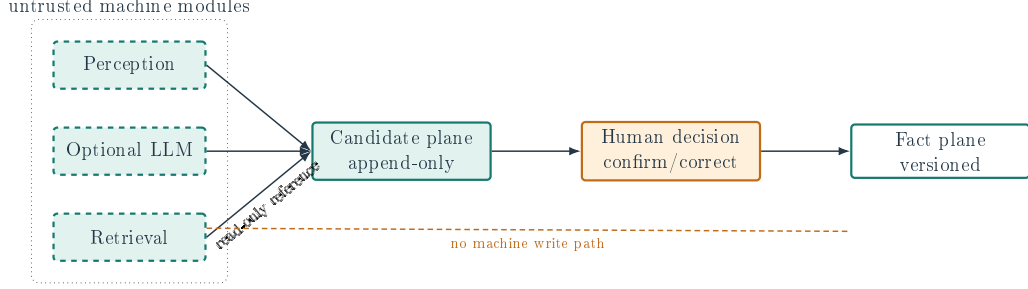


Figure 1: Trust model. Dashed machine components terminate at append-only candidates or read-only references; an attributable human decision is the only exposed path to a versioned fact.

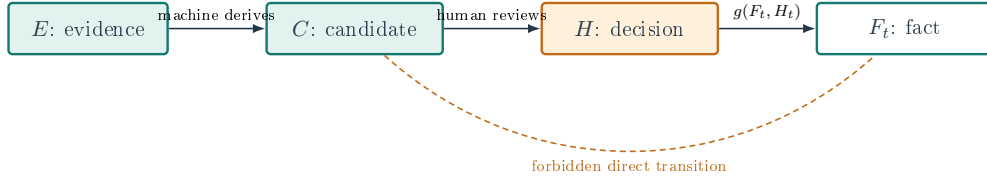


Figure 2: Allowed state transitions. Candidate-to-fact mutation without a human decision is forbidden by the application service boundary.

Equation 2 is an application-architecture property: no exposed machine service possesses a direct fact-writing operation. It is not a claim about protection against privileged database modification, compromised dependencies, or an adversary controlling the host.

For a recognition input x , a predictor returns a class $\hat{y}(x)$ and a selection score $s(x)$. Given threshold τ , the selection function is

$$a_\tau(x) = \mathbb{I}[s(x) \geq \tau]. \quad (3)$$

For a sample of n labeled cases, coverage and selective risk are

$$\widehat{\text{cov}}(\tau) = \frac{1}{n} \sum_{i=1}^n a_\tau(x_i), \quad (4)$$

$$\widehat{R}(\tau) = \frac{\sum_{i=1}^n a_\tau(x_i) \mathbb{I}[\hat{y}_i \neq y_i]}{\max(1, \sum_{i=1}^n a_\tau(x_i))}. \quad (5)$$

The erroneous auto-pass rate uses all cases in the denominator. These empirical measures characterize the recognition gate; they do not alter the authority rule. An accepted recognition is still a candidate, not a fact.

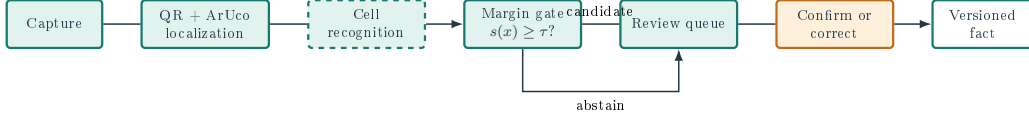


Figure 3: Operational pipeline. Recognition outcomes cross the threshold gate into the review queue but remain non-authoritative until a confirm/correct action.

The desired properties are: (P1) machine isolation, Equation 2; (P2) append-only candidate and fact histories; (P3) optimistic concurrency, so a stale decision cannot overwrite a newer version; (P4) content-addressed evidence and reverse trace from export to its supporting chain; and (P5) graceful operation when the optional AI adapter is disabled.

4. System and methodology

4.1. Template-guided perception

Each form template carries a QR payload identifying its template and version and four ArUco corner markers. The system decodes the QR, estimates a homography from detected corners, warps the image to a canonical canvas, and crops fields from versioned template coordinates. This converts a whole-page task into typed cell decisions and preserves the transformation parameters with the recognition attempt.

Digit crops are resized to 48×64 pixels and binarized with Otsu’s method. They are compared with ten rendered digit templates using normalized mean absolute difference. If $d_1(x)$ and $d_2(x)$ are the best and second-best distances, the selection score is the distance margin

$$s(x) = d_2(x) - d_1(x). \quad (6)$$

The candidate value is emitted only when $s(x) \geq \tau$; otherwise the recognizer emits an explicit abstention. OMR cells use an interior fill ratio: at most 0.15 is unchecked, at least 0.35 is checked, and the interval between them is rejected for review. These rules are intentionally inspectable rather than claimed as state of the art.

4.2. Candidate, decision, and fact services

Evidence is assigned a SHA-256 digest. Recognition attempts and AI reviews are stored independently from confirmed record versions. The optional

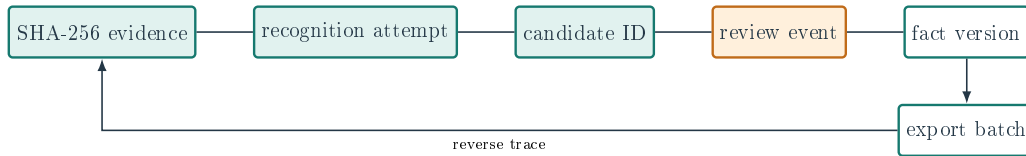


Figure 4: Evidence lineage from content-addressed input to export. Every arrow represents a persisted identifier used by the reverse-trace check.

AI adapter accepts structured context and must return schema-valid review JSON; its prompt contract states that it cannot modify business facts. The default adapter is disabled, returning an explicit `NOT_RUN` status. Local retrieval uses a dependency-free similarity index and exposes results only as references.

The review service accepts an expected record version, reviewer identifier, reason, candidate values, and evidence identifiers. It rejects a stale expected version, appends a new confirmed record on success, and emits a corresponding audit event. Exports query confirmed records only. Figure 4 shows the reverse-trace chain.

4.3. Implementation

The evaluated implementation uses Python 3.11, Streamlit, SQLite, SQLAlchemy, Alembic, OpenCV, scikit-learn, and an optional provider adapter. The application is organized into domain, application, adapter, and user-interface layers. It is a local single-machine prototype; it does not implement the FastAPI/React/PWA, authentication, or role-based-access features described in an earlier design concept. Reporting the implemented stack matters because the invariant is only as credible as the code path actually tested.

The system retains local evidence, database, and export files as one backup unit. This local-first operational choice supports recovery in a small-plant setting (Kleppmann et al., 2019), but it is not itself a security boundary. Deployment still requires host access control, encrypted backups, retention rules, and tested restore procedures.

5. Experimental protocol

5.1. Research questions

The evaluation addresses three questions. **RQ1** asks how risk-targeted abstention changes coverage and error exposure. **RQ2** asks whether injected

machine and workflow faults can change an existing fact through exposed application paths. **RQ3** asks whether core import–review–export operations, provenance, and concurrency checks remain functional without the AI adapter and across repeated executions.

5.2. *Synthetic recognition benchmark*

Production forms can contain employee identifiers, payroll-adjacent values, and commercially sensitive production data. They were therefore neither used in the public benchmark nor sent to external AI services. Instead, the package deterministically renders synthetic form cells with known labels. This decision makes the benchmark distributable and repeatable but limits ecological validity.

The benchmark contains 4,350 image rows. Seeds 101 and 202 define 1,740 calibration rows; disjoint seeds 303, 404, and 505 define 2,610 evaluation rows. The evaluation split contains 90 seed–font–digit base units, each observed under 29 clean or perturbed conditions. Consequently, rows are disjoint from calibration by seed but are not mutually independent within the evaluation split. Training data for the learned baseline use separate seeds. Perturbations comprise clean images plus Gaussian blur, Gaussian noise, brightness reduction, rotation, perspective distortion, JPEG compression, and occlusion at predefined severities. The implementation records every raw prediction, score, label, model, perturbation, severity, and seed.

Thresholds from 0 to 0.10 are swept in increments of 0.005. The operating threshold is selected *only on calibration rows* as the lowest threshold in the grid satisfying a target empirical selective risk of at most 1%, thereby retaining the highest calibration coverage among qualifying thresholds. The selected value is then frozen for evaluation. We report coverage, accepted accuracy, selective risk, erroneous auto-pass rate, and routing rate. The generated table also includes a descriptive row-level Wilson interval for accepted accuracy. Because that interval assumes independent Bernoulli trials whereas perturbations reuse base units, it is not treated as cluster-valid inferential evidence.

Three available recognition paths are compared: the Auto-Decte template matcher, the same template decision forced to predict whenever possible, and a HOG feature extractor with a linear support-vector machine. A Tesseract path is capability-gated. The executable was unavailable in the recorded environment, so no Tesseract score is reported.

5.3. Fault injection and resilience

Five application-path faults probe distinct boundaries: a wrong recognition candidate, a wrong language-model suggestion, an irrelevant retrieval result, a stale human-review replay, and duplicate evidence. A run passes only when the existing fact remains unchanged and an audit outcome is recorded; stale review and duplicate evidence must additionally raise their designated exceptions.

Resilience checks exercise import, human confirmation, export, reverse trace, duplicate rejection, stale-version rejection, and operation with AI disabled. Latency is measured over 30 repetitions after five warm-ups for both fresh-container (cold) and reused-container (warm) application flows. These sub-millisecond service timings exclude image decoding, browser rendering, storage-device variance, and human review, and therefore should not be interpreted as end-to-end user latency.

5.4. Reproducibility controls

The experiment code fixes seeds and dependency versions, emits JSON before figures or tables, and writes a manifest containing SHA-256 hashes, commands, environment metadata, source commit, and working-tree status. Paper tables are generated from the same JSON read by the plots. The final code quality gate comprises unit/integration tests, Ruff static checks, and mypy type checks. The evaluation is synthetic by design; a production claim would require a separate, access-controlled protocol with double transcription and adjudication.

6. Results

6.1. RQ1: selective recognition

Calibration selected $\tau = 0.08$ for the target empirical selective risk of at most 1%. On the seed-disjoint 2,610-row evaluation set, the gate accepted 292 predictions and routed 2,318 to review. Coverage was 11.19%, accepted accuracy was 100%, and no erroneous auto-pass was observed. The descriptive row-level Wilson 95% interval was 98.70%–100%, but it is optimistic as an inferential interval because the 29 conditions reuse 90 base units. Thus, the experiment observed zero errors among accepted rows; it does not establish a zero error probability.

Figure 5 shows the cost of this operating point. The full-coverage end of the curve has roughly 25% selective risk, while zero observed selective risk

Table 1: Seed-disjoint evaluation at the calibration-selected threshold $\tau = 0.08$. The Wilson interval is descriptive and assumes row independence.

Metric	Value
Coverage	11.2%
Accepted accuracy	100.0%
Selective risk	0.0%
Routing rate	88.8%
Accepted accuracy Wilson 95% CI	[98.7, 100.0]%

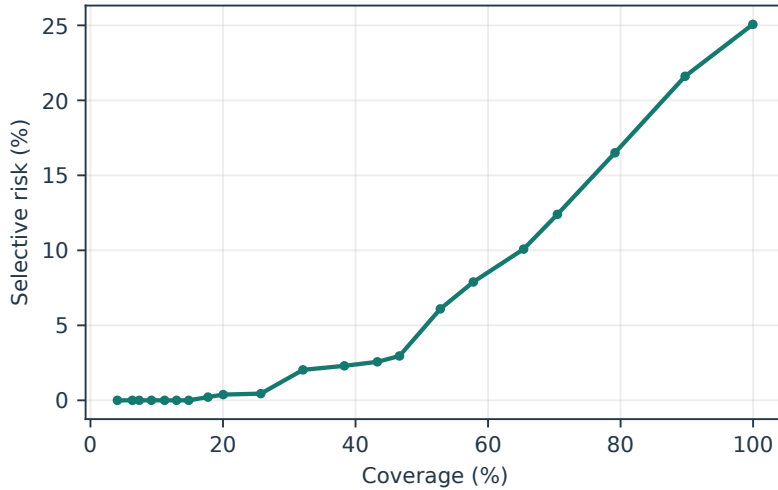


Figure 5: Evaluation coverage–risk trade-off over the threshold sweep. The dashed operating point was fixed from the calibration split before evaluation.

first occurs at $\tau = 0.07$ and increasingly reduces coverage. Risk control is therefore purchased with reviewer workload, not obtained for free.

The model comparison in Table 2 is important for interpretation. The template recognizer achieved 74.90% accuracy at effectively full coverage; forcing template prediction produced the same accuracy at exactly full coverage. HOG–linear-SVM achieved 99.81% (2,605/2,610) at full coverage. Consequently, Auto-Decte’s current recognizer is not competitive as a general classification contribution. The selective gate can isolate a small, high-confidence subset, while the stronger baseline indicates a direct path for improving coverage without changing the authority model.

Table 2: Recognition baselines on the seed-disjoint evaluation split.

Model	Accuracy (%)	Coverage (%)	Cases
auto-decte	74.9	100.0	2610
always-predict-template	74.9	100.0	2610
hog-linear-svm	99.8	100.0	2610

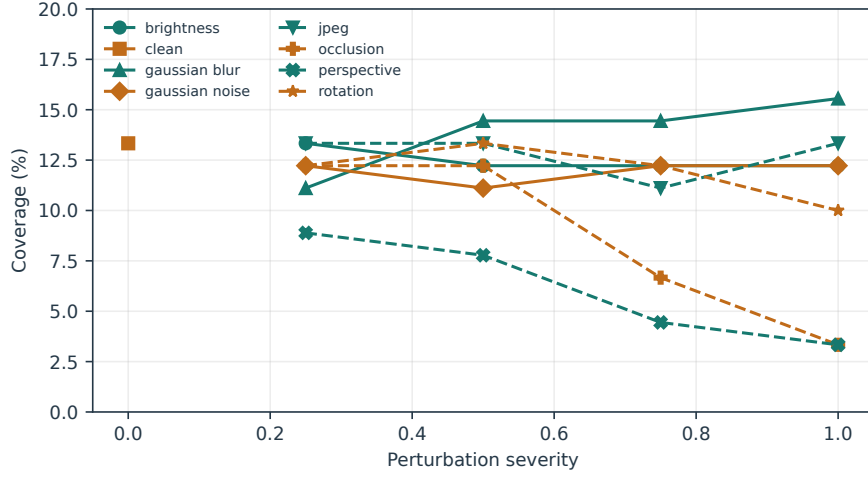


Figure 6: Coverage by perturbation and severity at $\tau = 0.08$. Each point aggregates the same 90 evaluation base units under one condition; accepted accuracy was 100% for every displayed group.

Across perturbations, accepted accuracy remained 100% at the selected threshold, but coverage declined under strong perspective distortion and occlusion to 3.33%. Figure 6 makes this asymmetric response visible: the conservative gate tends to abstain under severe degradation rather than preserve throughput.

6.2. RQ2: trust-boundary fault injection

All five injected faults left the pre-existing fact unchanged and yielded complete audit evidence (Table 3). Wrong recognition and wrong AI suggestions remained candidates; irrelevant retrieval remained a reference. The stale review replay raised `ConcurrentReviewError`, and duplicate evidence raised `DuplicateEvidenceError`. These are five controlled application-path cases, not a penetration test or exhaustive security proof.

Table 3: Trust-boundary fault outcomes. “Contained” requires an unchanged fact and a complete audit outcome.

Injected condition	Required mechanism	Outcome
Wrong recognition	Candidate isolation	Contained
Wrong AI suggestion	Separate AI-review record	Contained
Irrelevant retrieval	Read-only reference	Contained
Stale human replay	Optimistic concurrency	Rejected, contained
Duplicate evidence	Content-hash uniqueness	Rejected, contained

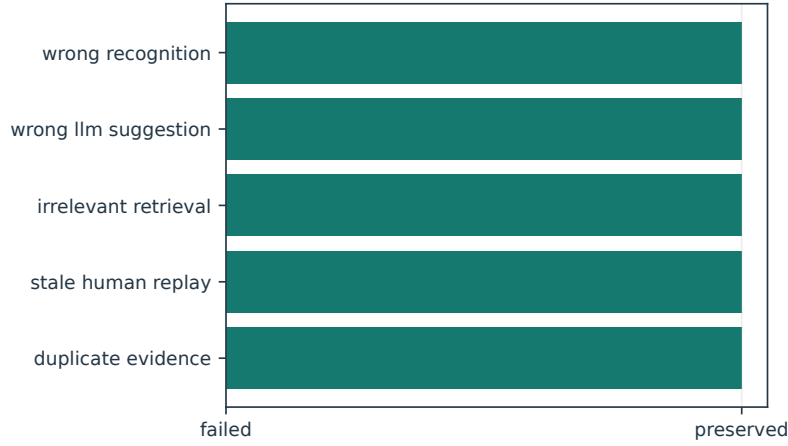


Figure 7: Fault-injection outcomes. A value of one denotes both an unchanged fact and complete audit evidence.

6.3. RQ3: resilience and reproducibility

The workflow completed with the AI adapter disabled, rejected duplicate evidence, rejected stale review, and returned a valid reverse trace. Across 30 measured repetitions, median cold service latency was 0.280 ms (IQR 0.071 ms; 95th percentile 0.517 ms), and median warm latency was 0.070 ms (IQR 0.014 ms; 95th percentile 0.129 ms). Figure 8 reports the distributions. These measurements validate low-overhead application services in the benchmark harness; they exclude perception and human interaction.

The final verification run passed 70 tests, Ruff checks, and mypy checks over 65 source files. The artifact manifest reports 15 hashed outputs and a clean tracked source state at generation time. These checks support repeatability of the stated synthetic results; they cannot substitute for external

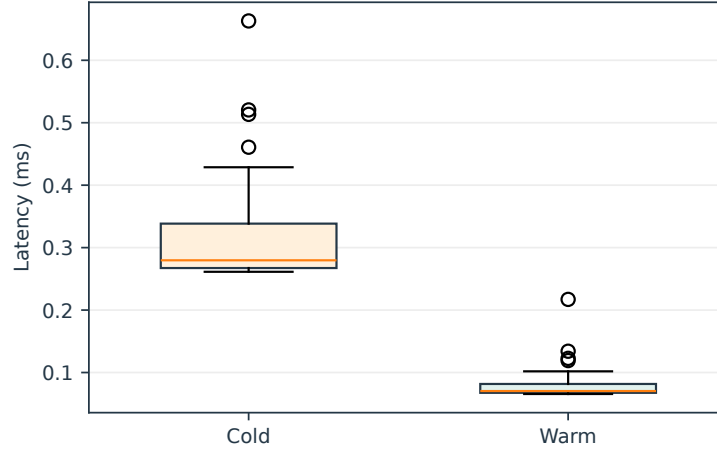


Figure 8: Application-service latency over 30 repetitions after five warm-ups. Values exclude image processing, UI rendering, and reviewer time.

replication or production validation.

7. Industrial context and intended use

The system was motivated by paper-based production records in a bamboo-processing setting, where form values can be adjacent to employee identity, output accounting, and payroll decisions. The intended user is a reviewer who receives a digitized form, inspects machine candidates alongside image evidence, confirms or corrects values, and exports only confirmed records. The prototype demonstrates that this workflow can be represented in software, but the paper does not claim a measured deployment effect on cost, time, accuracy, wages, or organizational outcomes.

The synthetic evaluation is a deliberate privacy and governance choice. Real records were excluded because public release could reveal personal or commercially sensitive information, and because sending them to third-party generative systems would require separate legal, contractual, and consent analysis. Synthetic forms preserve known labels, controlled perturbations, and distributable evidence. They do not reproduce handwriting variation, capture habits, paper wear, device diversity, or reviewer behavior in the plant.

For production adoption, the authority model should be embedded in a broader control system: least-privilege host access, encrypted storage and

backups, retention schedules, separation of reviewer and payroll roles where required, audit review, incident response, and periodic sampling of confirmed records. Candidate–fact separation reduces one route by which machine error becomes operational truth; it does not replace these organizational controls.

8. Discussion

The experiments expose two different notions of performance. Predictive performance asks how often the recognizer is correct. Governance performance asks whether an incorrect or irrelevant machine output can bypass an accountable decision. The HOG–SVM result shows that predictive performance can be substantially improved by replacing the current template matcher. The fault matrix shows that the authority boundary remains useful even when prediction is strong, because rare errors, model drift, irrelevant retrieval, and generative suggestions still require a controlled destination.

The selected threshold illustrates the practical tension in selective prediction. A 1% calibration risk target led to 11.19% evaluation coverage and 88.81% routing. This operating point is inappropriate if manual review capacity is scarce, even though its accepted subset was error-free in the sample. A deployment should therefore choose a threshold through an explicit cost model that values an unreviewed error, a review minute, and delayed processing. Calibration should be repeated after changes in forms, printers, cameras, or recognizers. The descriptive Wilson lower bound further cautions that 292 correct accepted rows do not imply perfect accuracy; dependence among perturbations makes formal uncertainty wider than the row-level calculation suggests.

The architecture is model-agnostic. A stronger classifier can supply candidates and confidence scores; a conformal or learned selector can replace the margin rule; and a different retrieval or language model can produce assistance. Each remains outside the fact-writing interface. This modularity separates model procurement from authority design and creates a stable audit question: which service can call the fact-transition operation?

The five fault injections should be viewed as executable design assertions. They provide stronger evidence than a diagram alone because they exercise concrete services and stored state. They are weaker than formal verification, adversarial security assessment, or coverage of every UI and migration path. Future assurance could add property-based state-machine tests, database

privilege separation, signed audit events, and independent red-team scenarios.

The most important next empirical study is not another synthetic recognition benchmark. It is a blinded, double-entered corpus of real forms, with adjudicated ground truth and subgroup reporting by template, device, writing style, and degradation. A second study should randomize review interfaces—recognition only, retrieval-assisted, and AI-assisted—and measure review time, correction accuracy, automation bias, and inappropriate suggestion acceptance. Those studies would test whether the architecture improves decisions rather than merely constraining software transitions.

9. Limitations and threats to validity

External validity. All quantitative recognition results use deterministic synthetic cells. They do not represent handwriting, Chinese text, signatures, paper damage, camera diversity, or the joint distribution of production forms. The privacy rationale for withholding production data is legitimate, but it does not turn synthetic evidence into field evidence.

Statistical validity. Threshold selection uses calibration seeds disjoint from evaluation seeds, reducing direct evaluation leakage. Nevertheless, calibration and evaluation arise from the same renderer and perturbation generator. More importantly, the 2,610 evaluation rows are 29 conditions applied to only 90 seed–font–digit base units. The row-level Wilson lower bound of 98.70% assumes independence and is therefore descriptive rather than a cluster-valid confidence bound. Zero observed accepted errors is not proof of perfect accuracy, and condition-level comparisons remain coarse.

Construct validity. The trust tests operationalize containment as an unchanged fact plus complete audit evidence. This captures the stated application invariant but not human susceptibility to misleading suggestions, correctness of a later human decision, confidentiality, availability, or resistance to host compromise. “Trust-constrained” should not be read as a general security certification.

Baseline validity. The HOG–SVM baseline is strong on this renderer and may benefit from the constrained visual regime. Conversely, Tesseract was not installed and was excluded rather than simulated. Broader OCR and document-model comparisons on real data remain necessary.

Implementation scope. The evaluated prototype is local and single-machine, with a Streamlit interface. It has not been load-tested for con-

current users and does not yet implement production authentication or role-based access. Sub-millisecond service timings are microbenchmarks, not end-to-end latency.

Reproducibility. Seeds, code, raw predictions, summary JSON, plots, tables, and hashes are supplied. Reproduction still depends on compatible OpenCV and system behavior, and independent execution has not yet been reported. The public package intentionally excludes sensitive plant records.

10. Conclusion

Auto-Decte treats machine outputs as evidence-linked candidates rather than authoritative facts. Within the exposed application services, perception, retrieval, and optional language-model review have no direct fact-writing path; a version-checked, attributable human decision is required. A deterministic synthetic evaluation showed how conservative abstention trades throughput for a high-accuracy accepted subset, while a strong HOG-SVM baseline demonstrated that recognition itself is not the paper’s novelty. Five injected faults were contained without changing existing facts, and the workflow remained functional with AI disabled.

The evidence supports the feasibility of the architecture and its executable invariants, not production effectiveness or universal safety. Real-form validation and controlled human-review studies are the necessary next steps. The central design lesson is portable: improve models when possible, but design the data path so that model confidence is never confused with authority.

Declarations

Data and code availability. The reproducible synthetic benchmark, raw predictions, summary artifacts, figure-generation code, and application source are included in the accompanying repository. Production forms are not included because they may contain employee identity, payroll-adjacent values, and commercially sensitive production information.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process. During the preparation of this work, the authors used DeepSeek Harness and OpenAI Codex to assist with drafting, content organization, language editing, and AI-assisted code development. The authors reviewed, edited, and verified the resulting text, code, analyses, references, and claims, and take full responsibility for the content

of the publication. No generative-AI-created or generative-AI-altered image is included in this submission; diagrams are author-controlled vector graphics and quantitative plots are generated programmatically from the reported experiment artifacts.

Funding, competing interests, and author contributions. These items are intentionally omitted from the double-anonymized manuscript and must be completed by the authors on the separate title page before submission.

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